Does Kilian secretly prefer DoRA adaptation?

A story of DoRA vs LoRA

Aryaman Sharma • Atindra Jayakar • Ishita Sinha[1]
[1]Computer and Information Science | Cornell University
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INTRODUCTION / BACKGROUND

PROBLEM

Full fine-tuning large language models is expensive, while LoRA is efficient but can still underperform compared to full fine-tuning on complex tasks.

MOTIVATION

We want to adapt large models with fewer trainable parameters while closing the gap between LoRA and full fine-tuning on downstream tasks.

GOAL AND HYPOTHESIS

We aimed to reproduce DoRA on LLaMA-3 8B for commonsense reasoning tasks.
Hypothesis: DoRA should outperform LoRA under the same rank and training budget.

APPROACH

DoRA separates weight adaptation into magnitude learning and low-rank directional updates.

$$\textbf{LoRA:}\quad W' = W_0 + \frac{\alpha}{r}BA$$

$$\textbf{Magnitude:}\quad m = \|W_0\|$$

$$\textbf{Direction:}\quad V = \frac{W_0}{\|W_0\|}$$

$$\textbf{DoRA Direction Update:}\quad V' = W_0 + \frac{\alpha}{r}BA$$

$$\textbf{DoRA:}\quad W' = m' \cdot \frac{W_0 + \frac{\alpha}{r}BA}{\|W_0 + \frac{\alpha}{r}BA\|}$$

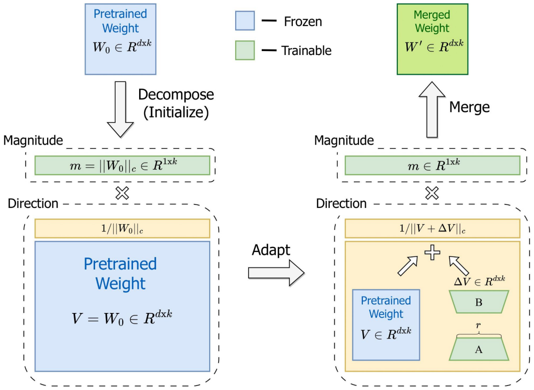


Figure 1: DoRA Weight decomposition adapted to existing LoRA architecture [1].

METHODOLOGY

MODEL

LLaMA-3.1-8B & DoRA adapters

DATASETS

BoolQ, PIQA, SIQA, HellaSwag, ARC-Easy, ARC-Challenge, Winogrande, OpenBookQA

DESIGN CHOICES & MODIFICATIONS

We froze the base model and injected DoRA adapters into LLaMA attention layers, with optional MLP adaptation

END TO END PIPELINE

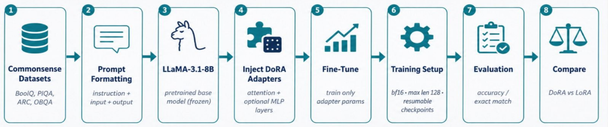
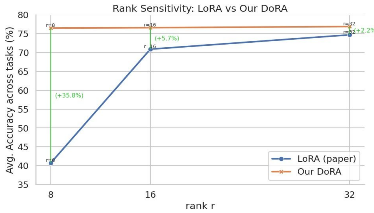


Figure 2: Step by step breakdown of training pipeline

ADDITIONAL EXPERIMENTS

Comparing LoRA and DoRA across different adapter ranks.

DoRA performed consistently across ranks, showing it is relatively rank-insensitive.



REFERENCES

[1] Liu, S.-Y., Wang, C.-Y., Yin, H., Molchanov, P., Wang, Y.-C. F., Cheng, K.-T., & Chen, M.-H. (2024). DoRA: Weight-Decomposed Low-Rank Adaptation. Proceedings of the 41st International Conference on Machine Learning, 235, 32100–32121. arXiv:2402.09353

RESULTS & CONCLUSION

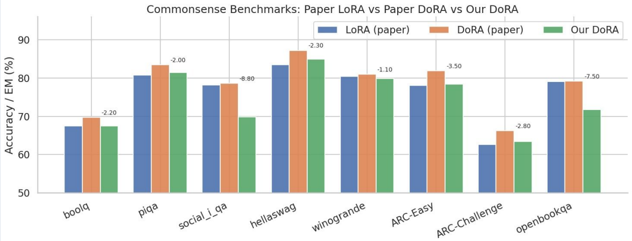


Figure 4: Performance on the Commonsense reasoning benchmark

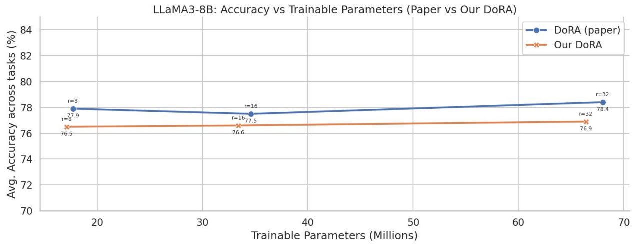


Figure 5: Comparison of Accuracy vs Trainable parameters

RESULTS

On BoolQ & PIQA our DoRA was within about 3-5% of paper DoRA.
The largest gap was on Social-I-QA, where our DoRA was around 8% within paper DoRA.
In the trainable-parameter experiment, our DoRA followed the same trend as the paper: accuracy stayed fairly stable as rank increased.

CONCLUSION

Overall, the results support the main claim that DoRA remains robust across adapter ranks and competitive with paper-scale results.

FUTURE WORK

Use an Instruct or Chat 8B model and save the custom adapter as a PEFT module for better inference time integration